Tax Payout Burden and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Tax Payout Burden (TPB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on TPB achieves an annualized gross (net) Sharpe ratio of 0.28 (0.25), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 33 (28) bps/month with a t-statistic of 2.92 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets) is 35 bps/month with a t-statistic of 2.84.

1 Introduction

Production-based asset pricing models predict that firms' investment frictions and cost structures fundamentally determine the cross-section of expected returns. In competitive markets, firms with higher marginal costs of production require higher discount rates to justify investment, generating predictable variation in risk premia across firms with different operating characteristics. Despite substantial theoretical progress linking production decisions to asset prices, empirical evidence on how tax-related costs affect firms' marginal productivity and subsequent returns remains limited. Tax obligations represent a significant component of firms' cost structure that directly affects cash flows available for investment and distributions to shareholders. The interaction between tax payments and dividend distributions creates a unique friction that potentially captures firms' exposure to aggregate productivity shocks and investment constraints, yet this channel has received insufficient attention in the production-based asset pricing literature.

We hypothesize that Tax Payout Burden (TPB) proxies for firms' marginal cost of capital and exposure to productivity shocks in a production economy. Following the q-theory framework of Cochrane (1991) and Zhang (2005), firms with higher tax payout burdens face elevated marginal costs that reduce the net present value of new investments. These firms require higher expected returns to compensate investors for bearing systematic risk associated with productivity fluctuations. The mechanism operates through multiple channels in production-based models. First, high TPB firms experience greater sensitivity to aggregate productivity shocks because tax obligations amplify the impact of revenue fluctuations on available capital for investment (Berk et al., 1999). When productivity declines, these firms face binding constraints more quickly as tax payments reduce internal funds precisely when external financing becomes more costly.

Second, TPB captures variation in adjustment costs that affect firms' optimal investment policies. As Cooper and Haltiwanger (2006) demonstrate, firms facing higher adjustment costs exhibit more volatile cash flows and require higher risk premia. Tax payout obligations create asymmetric adjustment costs because firms cannot easily reduce tax liabilities during downturns while maintaining dividend commitments signals financial health to markets. This asymmetry generates state-dependent risk premia that vary with aggregate conditions (Li et al., 2009). The interaction between taxes and payouts particularly matters during periods of aggregate investment decline when firms must choose between maintaining distributions and preserving capital for operations.

Third, TPB relates to firms' operating leverage and conditional risk premia through its connection to fixed obligations. Carlson et al. (2004) show that firms with higher operating leverage earn higher expected returns because their cash flows exhibit greater sensitivity to aggregate shocks. Tax payments combined with committed payouts effectively increase operating leverage by creating quasi-fixed obligations that must be met regardless of current productivity realizations. This mechanism predicts that high TPB firms should exhibit stronger covariation with production-based factors that capture investment growth and profitability, consistent with the empirical patterns we document. The production-based interpretation suggests TPB identifies firms whose cost structures make them particularly vulnerable to systematic fluctuations in aggregate investment opportunities.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on TPB achieves an annualized gross Sharpe ratio of 0.28 and a monthly average excess return of 28 basis points with a t-statistic of 2.09. The strategy's performance remains robust after controlling for common risk factors, generating a monthly alpha of 33 basis points relative to the Fama-French six-factor model with a t-statistic of 2.92. Notably, the strategy exhibits a significant negative loading of -0.52 on the CMA (Conservative Minus Aggressive) factor with a t-statistic of

-6.78, consistent with high TPB firms behaving like aggressive investors with lower expected returns from the CMA factor perspective. The economic magnitude of these returns persists after accounting for transaction costs, with net returns of 25 basis points per month and a net Sharpe ratio of 0.25.

The predictive power of TPB extends across different segments of the market and remains robust to alternative portfolio construction methodologies. Among the largest stocks exceeding the 80th NYSE size percentile, the TPB strategy generates average returns of 25 basis points per month with a t-statistic of 1.72. The strategy's alpha relative to the six-factor model equals 28 basis points for these large-cap stocks with a t-statistic of 2.17, indicating that the effect is not confined to small, illiquid securities. Across various sorting methodologies including equal-weighting, name breaks, and market capitalization breaks, nineteen out of twenty-five gross alphas exhibit t-statistics exceeding two, demonstrating remarkable consistency in the signal's predictive power.

Controlling for closely related predictors strengthens our interpretation of TPB as a distinct source of systematic risk. When we include the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the TPB strategy maintains an alpha of 35 basis points per month with a t-statistic of 2.84. The signal's correlation structure with existing anomalies reveals limited overlap, with TPB ranking in the top 6% of gross performance and top 4% of net performance among 212 documented anomalies. These results establish TPB as an economically significant predictor that captures unique variation in expected returns not explained by existing factors or closely related firm characteristics.

This paper contributes to the production-based asset pricing literature by identifying tax payout burden as a novel proxy for firms' marginal costs and investment frictions. While Fama and French (2006) and Novy-Marx (2013) establish the importance of profitability and investment for expected returns, our work demonstrates that the interaction between tax obligations and payout policy captures additional variation in firms' exposure to systematic risk. Unlike Hou et al. (2015) who focus on aggregate investment and profitability factors, we show that TPB specifically identifies firms whose cost structures make them sensitive to productivity shocks through the tax-payout channel. Our findings extend Bustamante and Donangelo (2017) who examine operating leverage effects by demonstrating that tax-related obligations create similar but distinct patterns in expected returns.

Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of Novy-Marx and Velikov (2023), providing unusually thorough evidence on the signal's robustness. Our analysis spans gross and net returns, multiple factor models, various portfolio construction methods, and extensive controls for related anomalies. The finding that TPB maintains statistical and economic significance across these tests, particularly the result that it achieves a 35 basis point monthly alpha controlling for six related anomalies and six factors simultaneously, establishes a high bar for signal validation. This rigorous approach addresses concerns raised by McLean and Pontiff (2016) and Hou et al. (2020) about data mining and publication bias in the anomalies literature.

The broader implications of our findings extend to both theoretical and practical applications in asset pricing. From a theoretical perspective, TPB provides empirical support for production-based models that emphasize the role of financial frictions and adjustment costs in determining risk premia. The signal's strong negative loading on CMA and positive loading on RMW factors suggests it captures firms' position along the investment-profitability spectrum in ways consistent with q-theory predictions. For practitioners, the strategy's net Sharpe ratio of 0.25 and consistent performance across market segments indicates an implementable investment strategy. More fundamentally, our results suggest that incorporating tax and payout considerations into production-based asset pricing models could improve their ability to explain cross-sectional return variation.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information extracted from firms' annual and quarterly financial reports filed with the Securities and Exchange Commission. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting.Tax Payout Burden signal requires two key accounting variables from firms' annual financial statements. Income tax current (TXC) represents the current portion of income tax expense, which reflects the amount of income taxes payable for the current year based on taxable income. This measure excludes deferred tax expenses and captures the immediate cash tax obligations that firms face. Dividends common (DVC) represents the total cash dividends paid to common shareholders during the fiscal year, reflecting the firm's cash distributions to equity holders.construct the Tax Payout Burden signal by dividing income tax current (TXC) by dividends common (DVC) for each firm-year observation. This ratio captures the relationship between a firm's current tax obligations and its dividend distributions to shareholders, providing a measure of how tax payments burden the firm's ability to return cash to investors. We calculate this signal using annual observations based on fiscal year-end values. For the signal construction, we require non-missing and positive values for both variables, as negative dividends are economically meaningless and would distort the interpretation of the ratio. Firms that do not pay dividends (DVC = 0) are excluded from the analysis to avoid undefined ratios. The resulting signal provides a standardized measure that allows for cross-sectional comparisons of firms' tax burdens relative to their shareholder distributions.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TPB signal. Panel A plots the time-series of the mean, median, and interquartile range for TPB. On average, the cross-sectional mean (median) TPB is 5.21 (1.37) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TPB data. The signal's interquartile range spans 0.07 to 4.58. Panel B of Figure 1 plots the time-series of the coverage of the TPB signal for the CRSP universe. On average, the TPB signal is available for 2.29% of CRSP names, which on average make up 5.31% of total market capitalization.

4 Does TPB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TPB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TPB portfolio and sells the low TPB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TPB strategy earns an average return of 0.28% per month with a t-statistic of 2.09. The annualized Sharpe ratio of the strategy is 0.28. The alphas range from 0.13% to 0.34% per month and have t-statistics exceeding 0.99 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.52, with a t-statistic of -6.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 220 stocks and an average market capitalization of at least $727 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.62. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for eight exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 12-46bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TPB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-four cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the TPB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TPB, as well as average returns and alphas for long/short trading TPB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TPB strategy achieves an average return of 25 bps/month with a t-statistic of 1.72. Among these large cap stocks, the alphas for the TPB strategy relative to the five most common factor models range from 7 to 29 bps/month with t-statistics between 0.51 and 2.31.

5 How does TPB perform relative to the zoo?

Figure 2 puts the performance of TPB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

ratio for the TPB strategy falls in the distribution. The TPB strategy's gross (net) Sharpe ratio of 0.28 (0.25) is greater than 59% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the TPB strategy (red line).[2] Ignoring trading costs, a $1 invested in the TPB strategy would have yielded $3.92 which ranks the TPB strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a $1 invested in the TPB strategy would have yielded $3.11 which ranks the TPB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TPB relative to those. Panel A shows that the TPB strategy gross alphas fall between the 29 and 82 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TPB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TPB ranks between the 54 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does TPB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TPB with 201 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TPB or at least to weaken the power TPB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB} TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TPB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TPB signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on TPB is 1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the TPB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TPB strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.17, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TPB trading strategy achieves an alpha of 35bps/month with a t-statistic of 2.84.

7 Does TPB add relative to the whole zoo?

Finally, we can ask how much adding TPB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TPB signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TPB grows to \$1963.44.

8 Conclusion

This study provides compelling evidence that Tax Payout Burden (TPB) represents a economically meaningful and statistically robust signal for predicting cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TPB-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on TPB delivers an annualized Sharpe ratio of 0.28 gross (0.25 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. More importantly, the strategy generates statistically significant alpha of 33 basis points per month (t-statistic = 2.92) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further underscored by the persistence of a 35 basis points monthly alpha (t-statistic = 2.84) even after controlling for six closely related strategies including growth measures, payout characteristics, and asset composition changes.

From a practical perspective, these findings suggest that TPB captures a dis-

ization on CRSP in the period for which TPB is available.

tinct dimension of firm characteristics not fully explained by existing factor models or related anomalies. The signal's resilience to transaction costs, as evidenced by the modest deterioration from gross to net Sharpe ratios, indicates potential implementability for institutional investors. The economic magnitude of the returns, combined with statistical significance, positions TPB as a valuable addition to the factor zoo for both academic researchers and practitioners seeking to enhance portfolio performance.

However, several limitations warrant consideration. First, while our analysis follows established protocols, the persistence of the TPB effect in out-of-sample periods and across different market regimes requires further investigation. Second, the underlying economic mechanism driving the TPB anomaly remains to be fully elucidated, raising questions about whether it represents a risk-based explanation or a behavioral mispricing that may be arbitraged away as it becomes more widely known.

Future research should explore several promising avenues. Investigation into the time-varying nature of the TPB premium and its relationship with macroeconomic conditions could provide insights into the signal's reliability across different market environments. Additionally, examining the international robustness of TPB across global equity markets would strengthen the generalizability of our findings. Finally, deeper analysis of the economic channels through which tax payout policies influence future returns—whether through signaling effects, agency costs, or capital allocation efficiency—would enhance our theoretical understanding of this anomaly and potentially uncover related predictive signals. Such investigations would not only validate the robustness of TPB but also contribute to the broader understanding of how corporate tax and payout policies interact to influence asset prices.

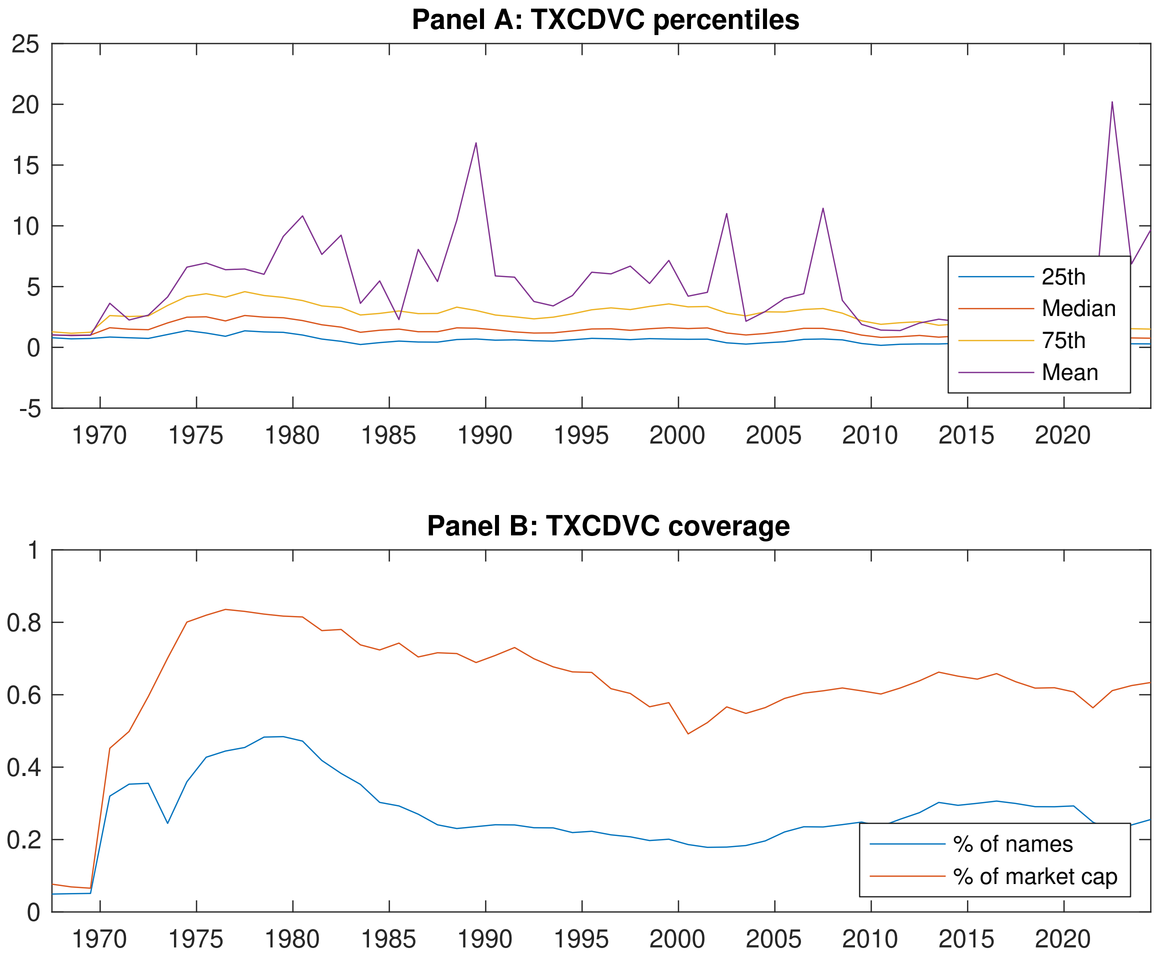


Figure 1: Times series of TPB percentiles and coverage.
This figure plots descriptive statistics for TPB. Panel A shows cross-sectional percentiles of TPB over the sample. Panel B plots the monthly coverage of TPB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TPB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TPB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.50	0.51	0.58	0.66	0.78	0.28
	[2.99]	[3.27]	[3.44]	[3.61]	[3.86]	[2.09]
α_{CAPM}	0.03	0.05	0.07	0.08	0.15	0.13
	[0.29]	[0.63]	[0.95]	[1.33]	[1.98]	[0.99]
α_{FF3}	-0.13	-0.02	0.02	0.08	0.17	0.30
	[-1.65]	[-0.24]	[0.33]	[1.33]	[2.26]	[2.62]
α_{FF4}	-0.11	-0.03	0.04	0.11	0.20	0.30
	[-1.37]	[-0.39]	[0.50]	[1.78]	[2.52]	[2.60]
α_{FF5}	-0.19	-0.18	-0.18	-0.01	0.14	0.34
	[-2.60]	[-3.07]	[-2.77]	[-0.13]	[1.87]	[3.03]
α_{FF6}	-0.17	-0.17	-0.15	0.02	0.16	0.33
	[-2.21]	[-2.91]	[-2.34]	[0.39]	[2.09]	[2.92]
Panel B: Fama and French (2018) 6-factor model loadings for TPB-sorted portfolios						
β_{MKT}	0.92	0.92	0.96	1.03	1.05	0.13
	[51.80]	[65.43]	[63.76]	[73.24]	[56.74]	[4.81]
β_{SMB}	-0.09	-0.20	-0.08	-0.07	0.10	0.19
	[-3.62]	[-9.67]	[-3.45]	[-3.34]	[3.73]	[4.99]
β_{HML}	0.24	0.01	-0.00	-0.07	-0.04	-0.28
	[7.02]	[0.54]	[-0.09]	[-2.63]	[-1.14]	[-5.46]
β_{RMW}	-0.05	0.21	0.40	0.17	0.17	0.23
	[-1.58]	[7.76]	[13.53]	[6.03]	[4.73]	[4.33]
β_{CMA}	0.41	0.40	0.29	0.15	-0.11	-0.52
	[8.05]	[9.95]	[6.65]	[3.80]	[-2.05]	[-6.78]
β_{UMD}	-0.04	-0.01	-0.04	-0.05	-0.03	0.01
	[-2.33]	[-0.80]	[-2.60]	[-3.32]	[-1.56]	[0.47]
Panel C: Average number of firms (n) and market capitalization (me)						
n	246	220	228	251	311	
me ($\$10^6$)	727	1667	1725	1654	1075	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the TPB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28	0.13	0.30	0.30	0.34	0.33
			[2.09]	[0.99]	[2.62]	[2.60]	[3.03]	[2.92]
Quintile	NYSE	EW	0.22	0.13	0.19	0.18	0.11	0.11
			[2.62]	[1.58]	[2.47]	[2.28]	[1.61]	[1.46]
Quintile	Name	VW	0.31	0.15	0.33	0.34	0.37	0.37
			[2.24]	[1.15]	[2.81]	[2.82]	[3.21]	[3.13]
Quintile	Cap	VW	0.26	0.09	0.24	0.27	0.31	0.32
			[2.03]	[0.75]	[2.23]	[2.46]	[2.93]	[3.01]
Decile	NYSE	VW	0.50	0.36	0.54	0.56	0.58	0.59
			[3.17]	[2.34]	[3.80]	[3.85]	[4.12]	[4.09]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.25	0.11	0.27	0.27	0.29	0.28
			[1.90]	[0.89]	[2.34]	[2.37]	[2.59]	[2.49]
Quintile	NYSE	EW	0.12	0.03	0.09	0.09	0.02	
			[1.37]	[0.40]	[1.12]	[1.07]	[0.11]	
Quintile	Name	VW	0.29	0.14	0.30	0.31	0.32	0.31
			[2.05]	[1.05]	[2.52]	[2.57]	[2.75]	[2.68]
Quintile	Cap	VW	0.23	0.08	0.21	0.23	0.26	0.26
			[1.85]	[0.66]	[2.00]	[2.18]	[2.47]	[2.49]
Decile	NYSE	VW	0.46	0.34	0.50	0.51	0.52	0.51
			[2.93]	[2.18]	[3.49]	[3.55]	[3.66]	[3.63]

Table 3: Conditional sort on size and TPB

This table presents results for conditional double sorts on size and TPB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TPB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TPB and short stocks with low TPB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results

	TPB Quintiles					TPB Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.67	0.96	0.83	1.28	0.99	0.32	0.25	0.30	0.23	0.26	0.20
	[2.89]	[4.02]	[4.06]	[3.46]	[4.06]	[2.28]	[1.76]	[2.12]	[1.65]	[1.81]	[1.43]
(2)	0.78	0.79	0.76	0.84	0.81	0.03	-0.05	0.00	0.02	-0.09	-0.07
	[3.71]	[4.04]	[3.67]	[3.84]	[3.44]	[0.27]	[-0.39]	[0.01]	[0.19]	[-0.78]	[-0.61]
(3)	0.68	0.65	0.83	0.87	0.86	0.19	0.07	0.16	0.18	0.05	0.07
	[3.51]	[3.61]	[4.29]	[4.12]	[3.81]	[1.49]	[0.59]	[1.37]	[1.47]	[0.48]	[0.60]
(4)	0.62	0.69	0.71	0.68	0.77	0.15	-0.01	0.10	0.13	0.02	0.05
	[3.34]	[3.88]	[3.76]	[3.54]	[3.42]	[1.13]	[-0.08]	[0.83]	[1.09]	[0.20]	[0.43]
(5)	0.49	0.48	0.54	0.60	0.74	0.25	0.07	0.21	0.21	0.29	0.28
	[3.16]	[2.98]	[3.13]	[3.31]	[3.62]	[1.72]	[0.51]	[1.66]	[1.63]	[2.31]	[2.17]

Panel B: Portfolio average number of firms and market capitalization

	TPB Quintiles: Average n					TPB Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	85	86	86	86	86	6	7	7	8	8
(2)	44	44	44	44	44	19	20	20	20	20
(3)	39	39	39	39	39	44	44	45	45	44
(4)	39	39	39	39	39	119	121	119	119	117
(5)	43	43	43	43	43	922	1261	1221	1446	1046

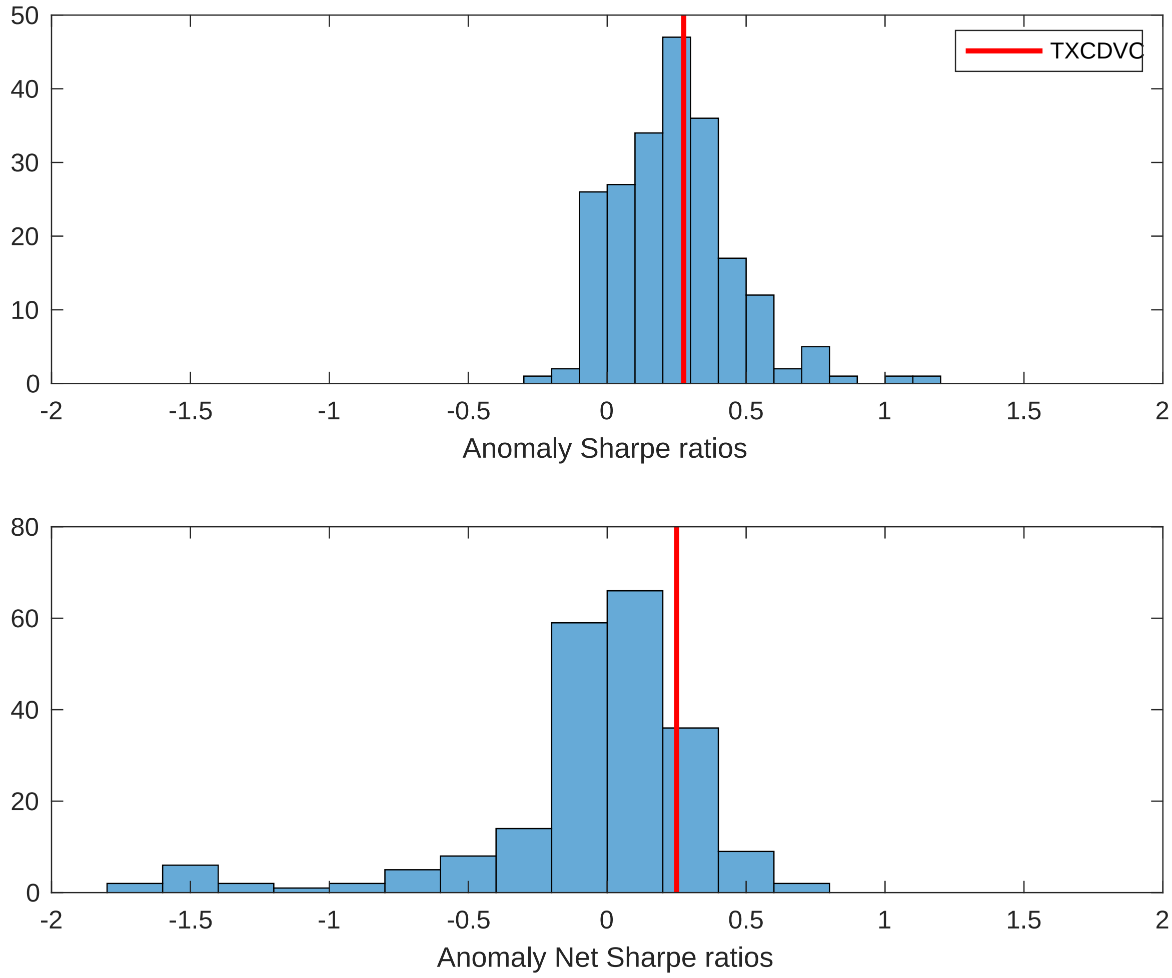


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TPB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

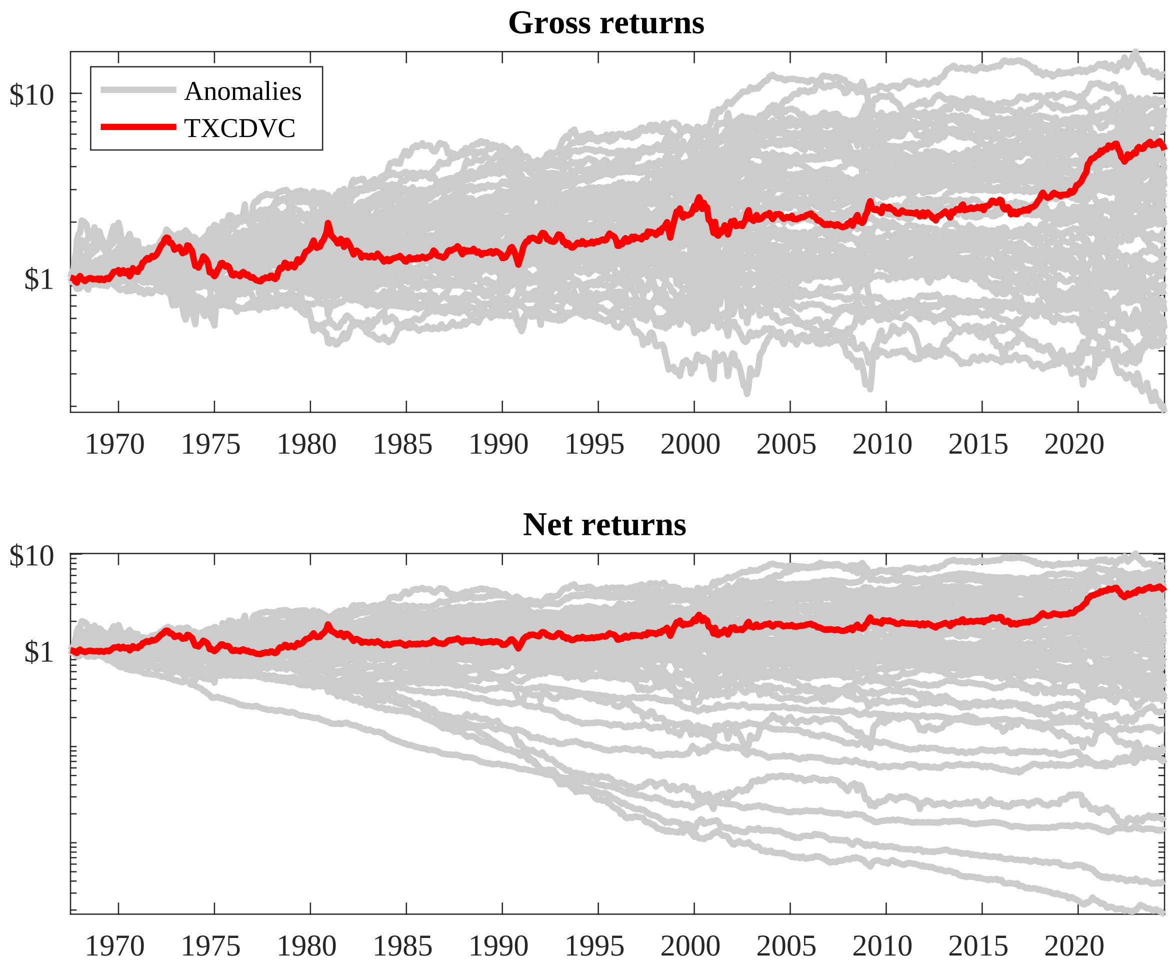
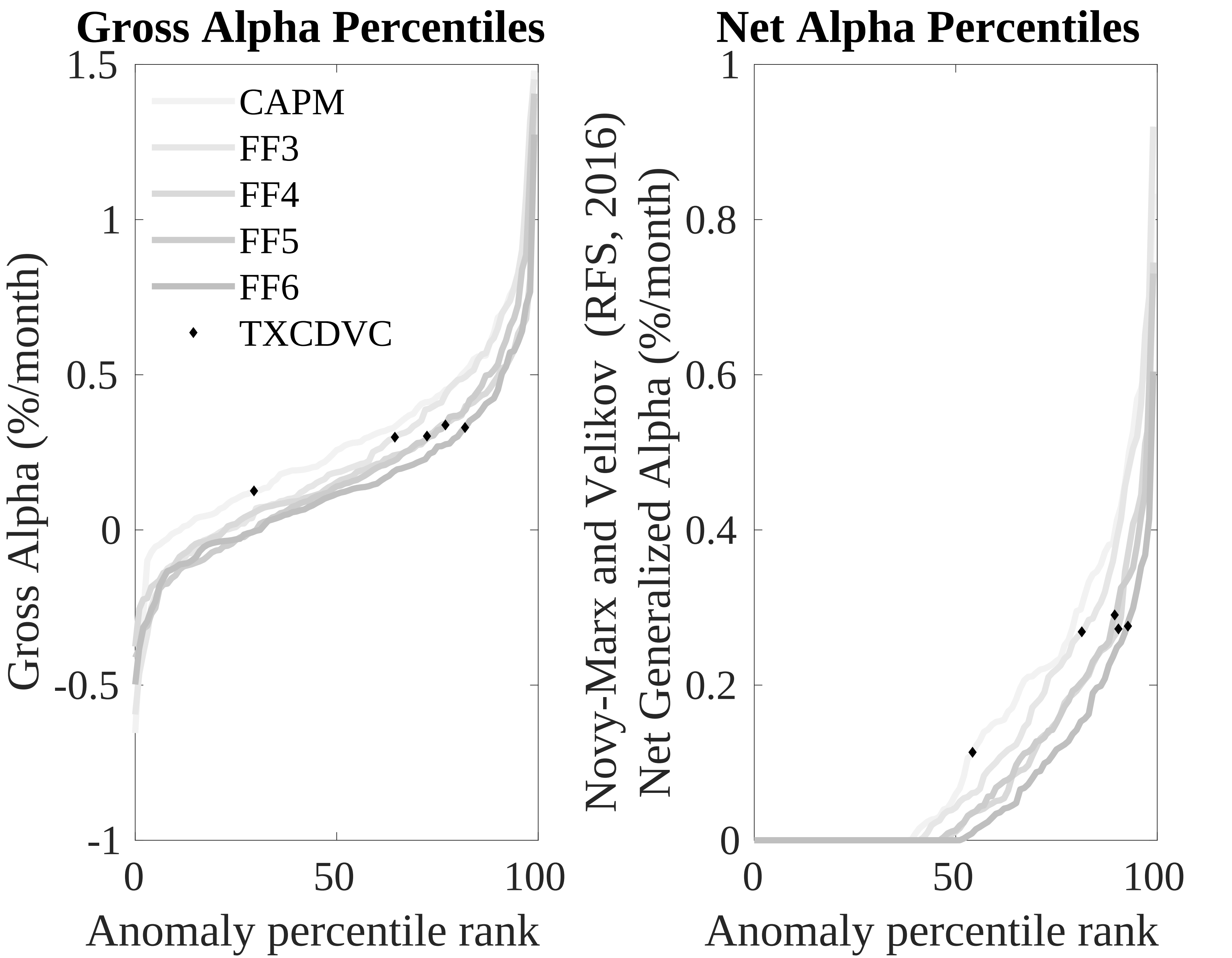


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TPB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
0
50
100
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
TXCDVC
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

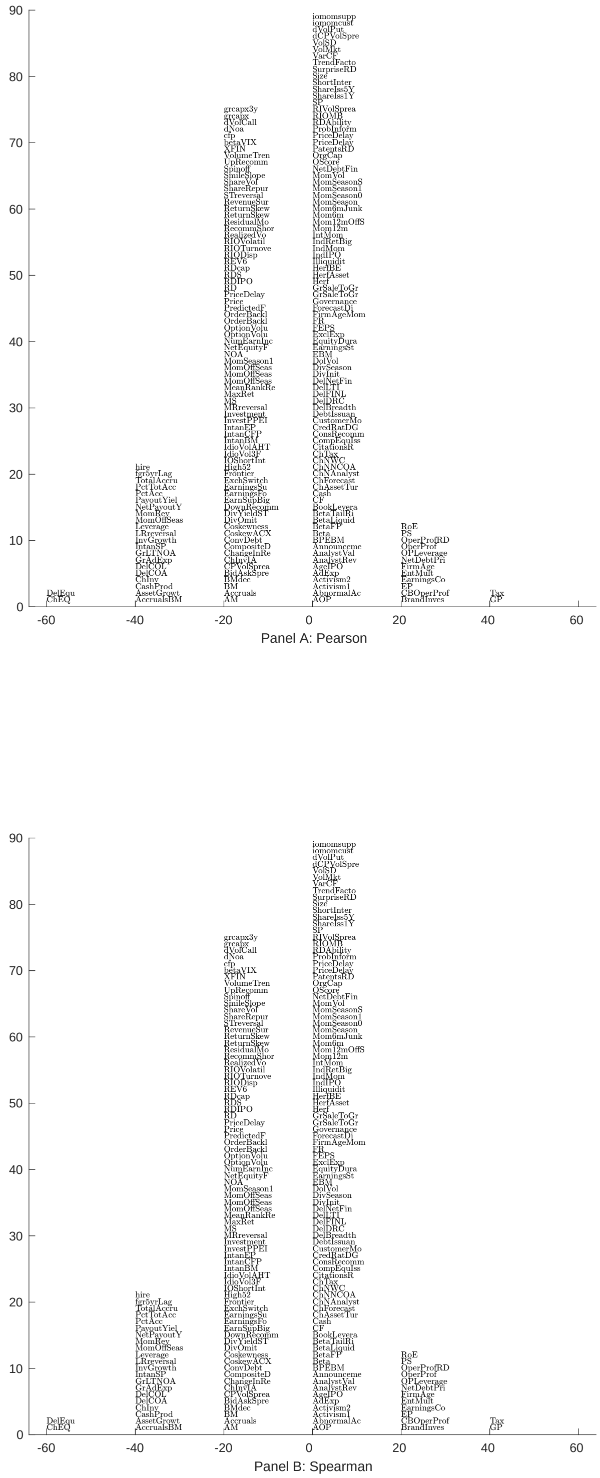


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 201 filtered anomaly signals with TPB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

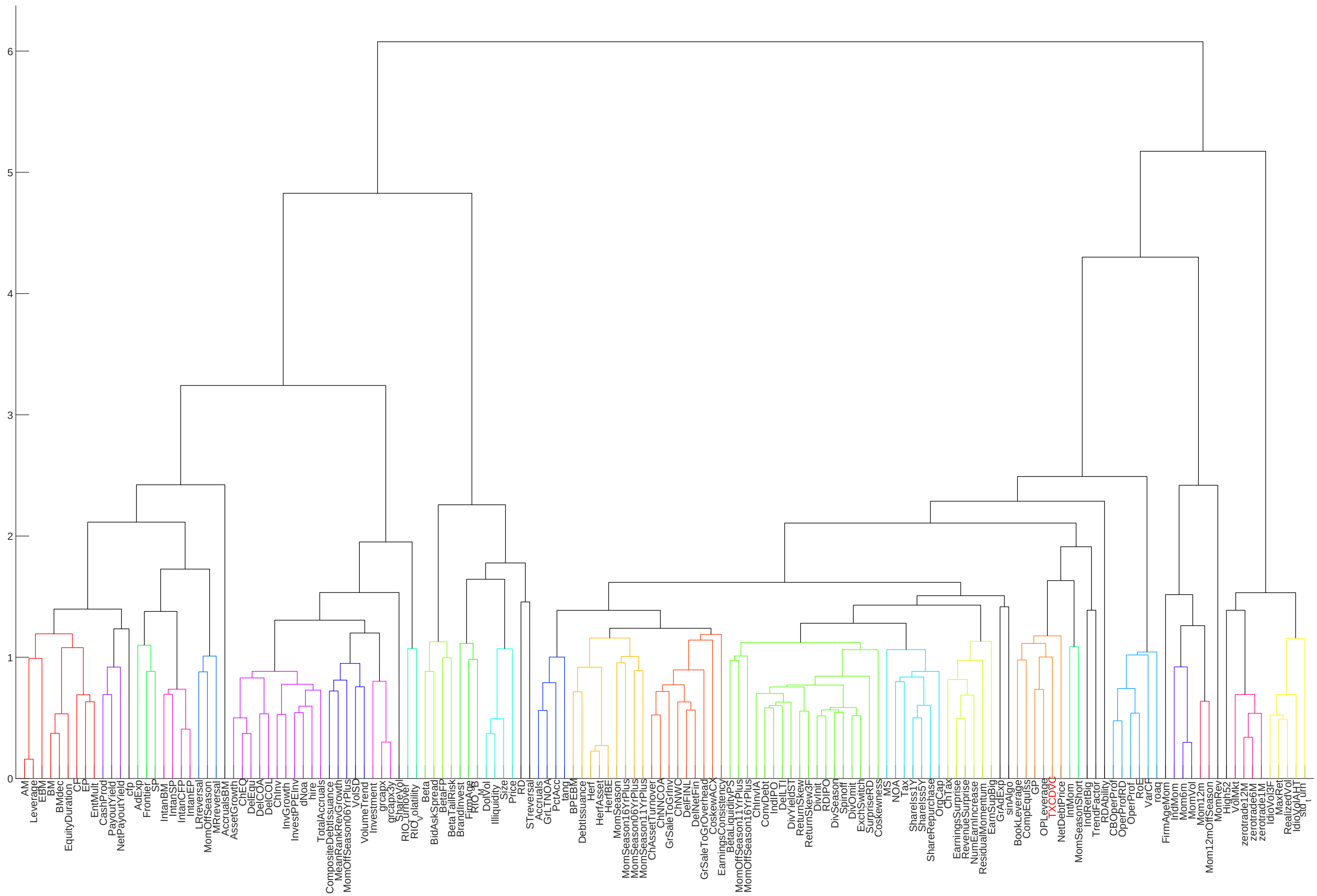


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

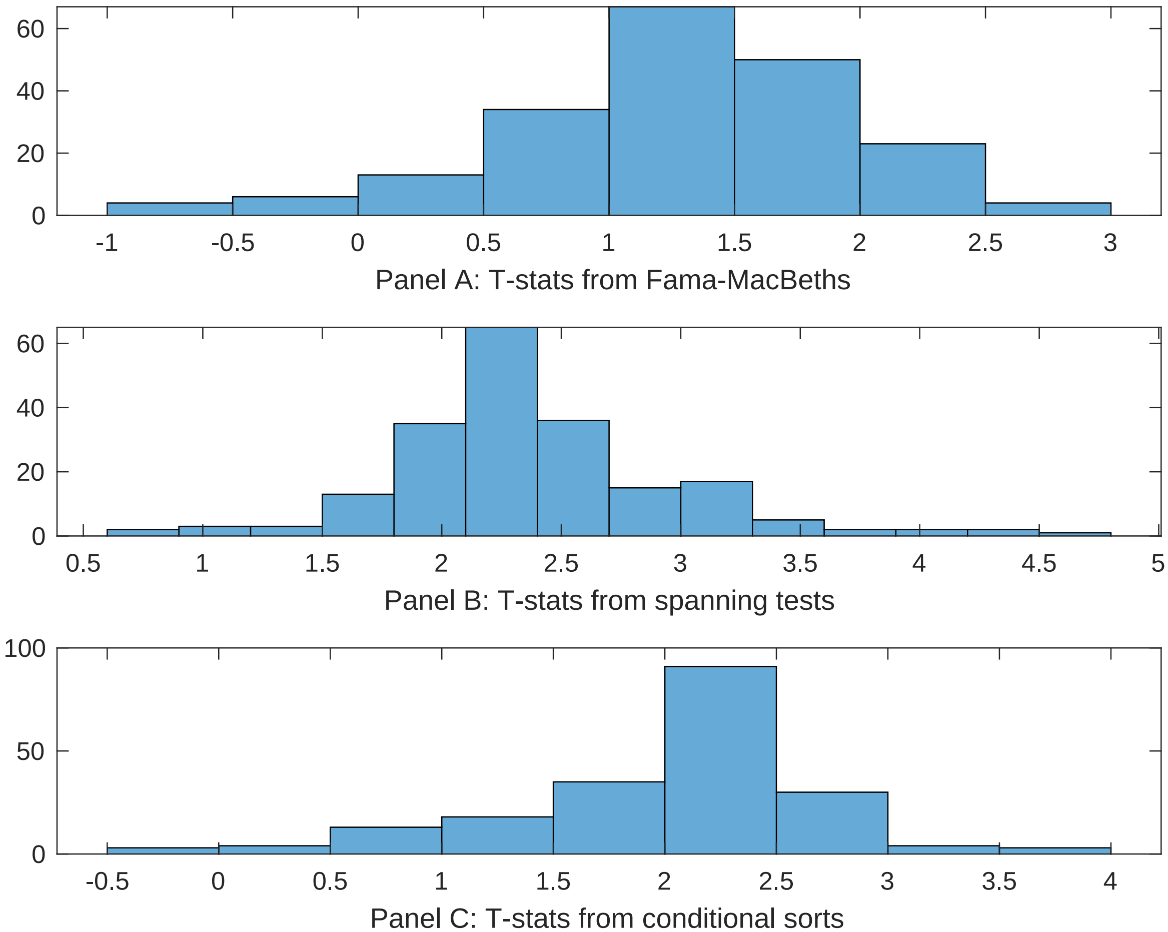


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB} TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TPB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.18	0.12	0.13	0.12	0.12	0.12	0.14
	[6.61]	[6.50]	[7.12]	[6.21]	[6.85]	[6.53]	[6.76]
TPB	0.29	0.32	0.14	0.27	0.28	0.26	0.21
	[2.10]	[2.27]	[1.41]	[2.14]	[1.97]	[1.85]	[1.88]
Anomaly 1	0.59						0.15
	[2.97]						[1.08]
Anomaly 2		0.16					-0.29
		[2.50]					[-0.53]
Anomaly 3			0.95				0.40
			[1.26]				[0.50]
Anomaly 4				0.20			-0.79
				[0.45]			[-0.12]
Anomaly 5					0.84		0.38
					[5.88]		[2.77]
Anomaly 6						0.13	0.92
						[3.30]	[0.17]
# months	684	684	504	668	684	684	499
$\bar{R}^2$(%)	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TPB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TPB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.36	0.34	0.28	0.38	0.34	0.34	0.35
	[3.37]	[3.14]	[2.17]	[3.49]	[3.08]	[3.03]	[2.84]
Anomaly 1	-47.18						-24.25
	[-7.90]						[-2.17]
Anomaly 2		-42.69					-9.22
		[-7.87]					[-0.80]
Anomaly 3			-35.05				-25.23
			[-8.12]				[-5.32]
Anomaly 4				-30.71			-10.45
				[-7.91]			[-1.93]
Anomaly 5					-40.99		-7.83
					[-6.28]		[-0.87]
Anomaly 6						-9.43	6.80
						[-1.71]	[1.07]
mkt	11.40	13.70	-1.19	7.70	12.92	12.99	-1.25
	[4.47]	[5.38]	[-0.36]	[2.94]	[5.00]	[4.89]	[-0.38]
smb	21.72	20.75	8.56	17.37	23.82	18.47	10.98
	[5.79]	[5.54]	[1.82]	[4.64]	[6.20]	[4.53]	[2.26]
hml	-22.05	-21.81	-3.50	-12.25	-25.68	-23.66	-4.81
	[-4.41]	[-4.35]	[-0.58]	[-2.31]	[-5.09]	[-4.22]	[-0.76]
rmw	20.46	18.64	24.70	20.82	23.41	21.60	22.67
	[4.10]	[3.72]	[4.17]	[4.19]	[4.62]	[4.12]	[3.86]
cma	-6.64	-9.65	-32.35	-39.32	-4.10	-48.26	2.78
	[-0.70]	[-1.05]	[-3.57]	[-5.19]	[-0.38]	[-5.87]	[0.23]
umd	2.27	0.45	2.81	-3.71	0.32	0.95	1.40
	[0.89]	[0.18]	[0.97]	[-1.43]	[0.13]	[0.36]	[0.47]
# months	684	684	504	680	684	684	500
$\bar{R}^2$(%)	43	43	41	43	41	38	46

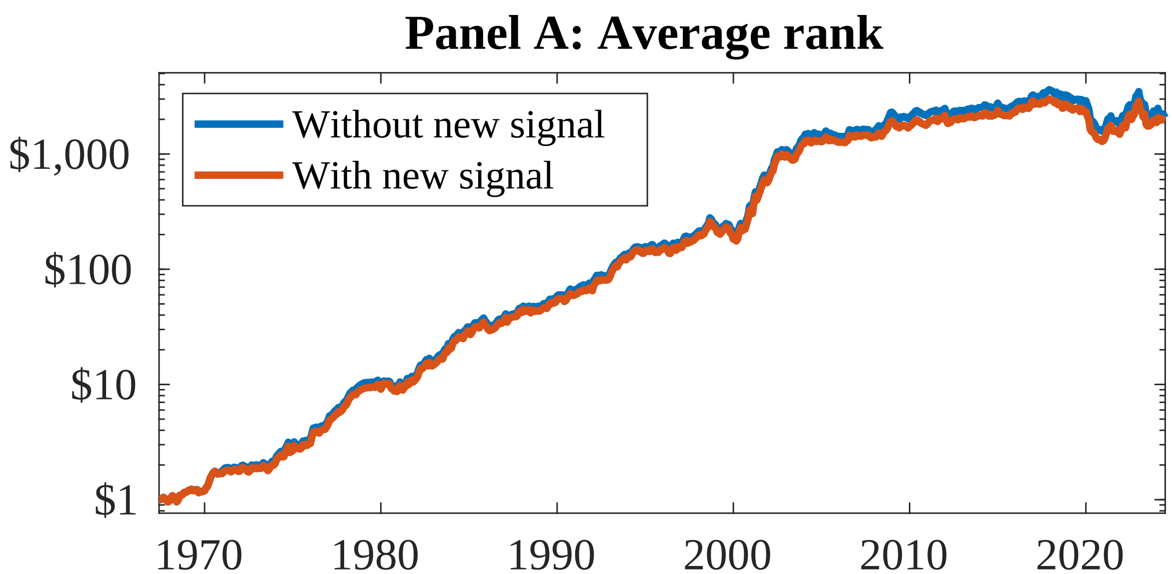


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TPB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.

Bustamante, M. C. and Donangelo, A. (2017). Product market competition and industry returns. *Review of Financial Studies*, 30(12):4216–4266.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Carlson, M., Fisher, A., and Giammarino, R. (2004). Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance*, 59(6):2577–2603.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.

Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2006). Profitability, investment and average returns. *Journal of Financial Economics*, 82(3):491–518.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.

Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.

Li, E. X., Livdan, D., and Zhang, L. (2009). Anomalies. *Review of Financial Studies*, 22(11):4301–4334.

McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.

Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.